



Thermal Trace

AI COLD-CHAIN MANAGEMENT • FOOD-LOSS PREVENTION

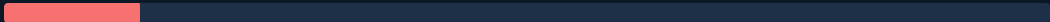
Know what a failing fridge is costing **before the gate does.**

It watches food in transit, measures how much safe life a failure has cost, and chooses the action that loses the least of it.

SAFE MAX 4 °C

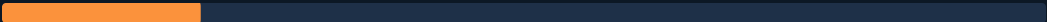
A third of our food never gets eaten.

33%



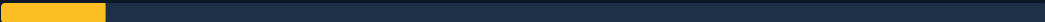
of food is **lost** between harvest and retail.

33%



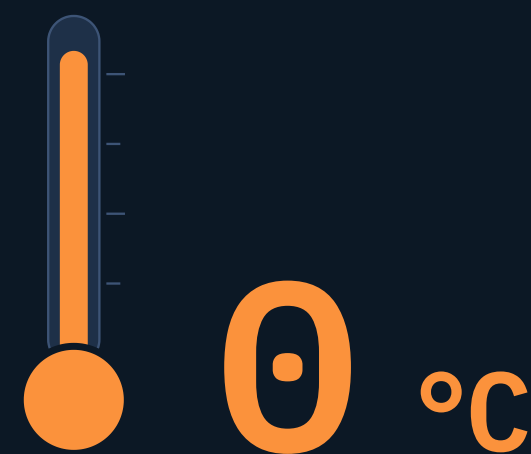
more is **wasted** after it reaches retail and homes.

33-33%



of global greenhouse-gas emissions come from food loss and waste.

Qatar imports most of its fresh food across a summer that passes 45 °C.



A fridge that fails at noon can spoil a load in hours.

NATIONAL FOOD SECURITY STRATEGY 2030

Food waste

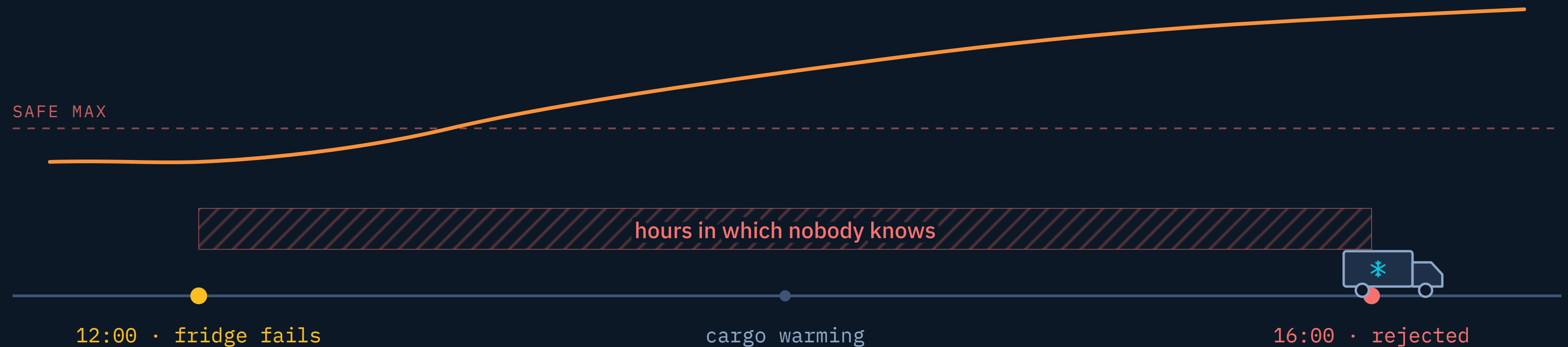
–50%

Food loss

–30%

Targets set by the State of Qatar. Thermal Trace works on the loss in transit.

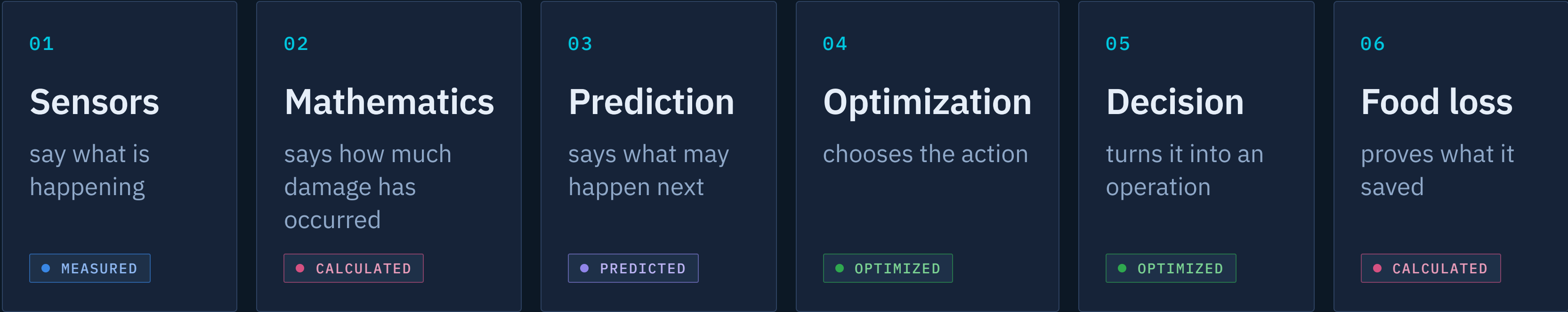
Nobody finds out until the load is **rejected at the gate.**



By then the food is gone, and so is any chance to send it somewhere it could still be sold.

Physics first. AI where it earns its place.

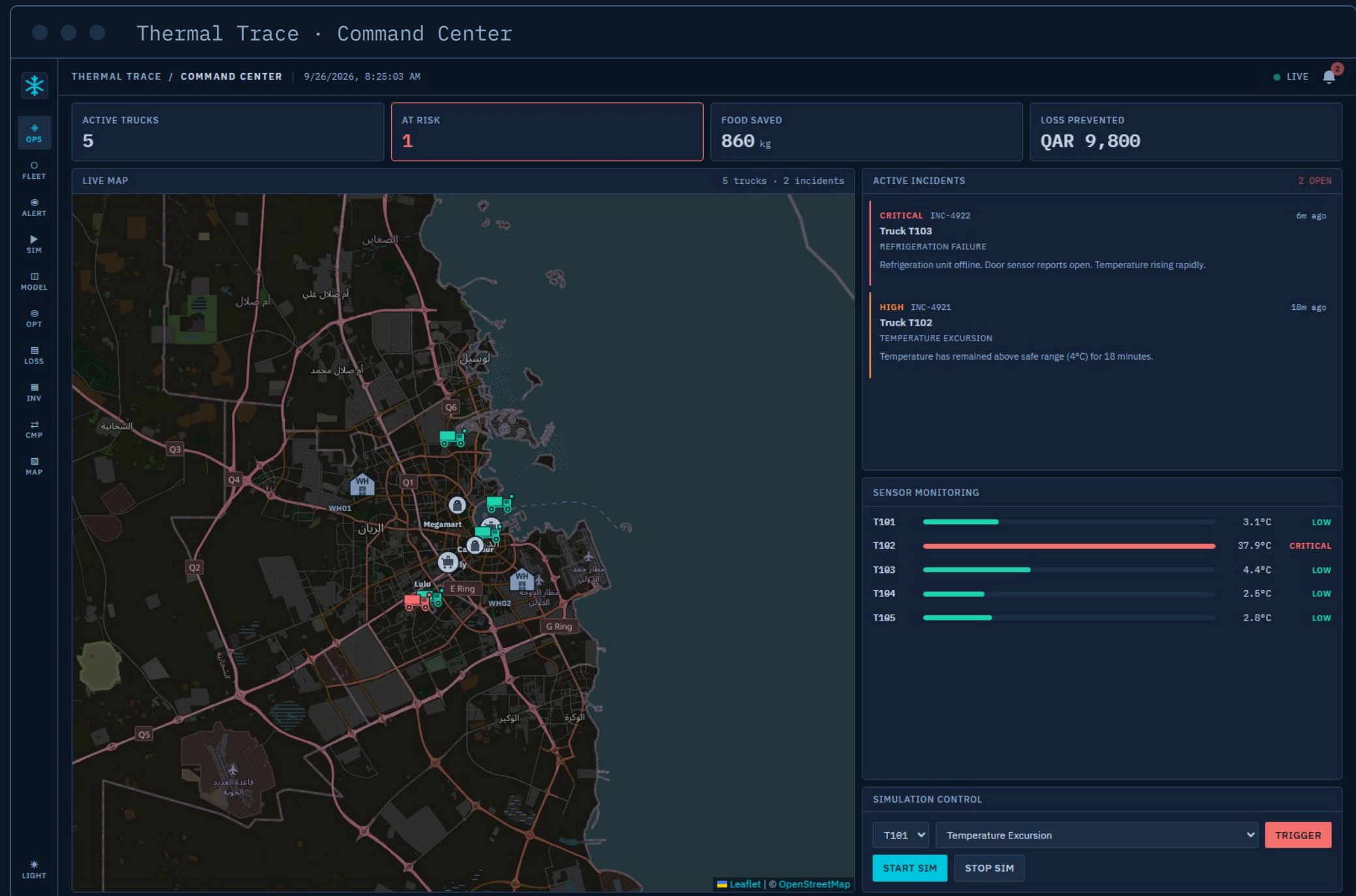
Sensors say what is happening, the maths says how much damage is done, and the optimizer chooses the action that loses the least food.



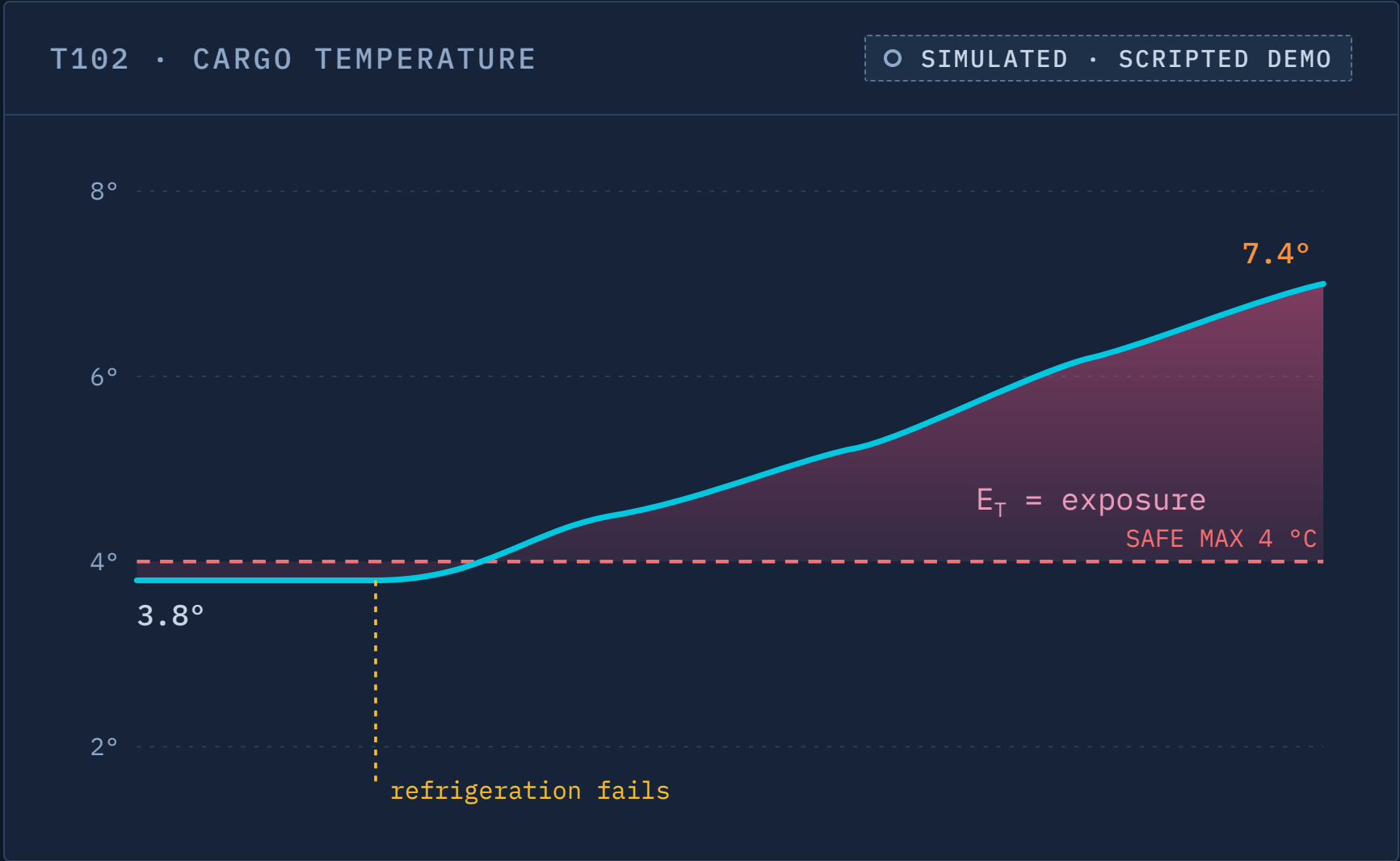
At the moment of failure, not at the gate.

A live map of every truck, with an incident the instant a temperature leaves the safe band and stays there.

- A decision surface, not a dashboard of charts.



Damage is **measured**, not guessed.



THERMAL EXPOSURE • °C·MIN

$$E_T = \sum \max(0, T_i - T_{\text{safe}}) \cdot \Delta t$$

DETERIORATION RATE • ARRHENIUS

$$k(T) = k_{\text{ref}} \cdot e^{(E_a/R)(1/T_{\text{ideal}} - 1/T)}$$

SHELF LIFE SPENT • REMAINING

$$D = \sum k(T_i) \cdot \Delta t_i \rightarrow L \cdot (1 - D)$$

● CALCULATED

Deterministic Python. No model touches these numbers.

Two AI layers. Neither may invent a number.

SYSTEM 1 • LOCAL • OFFLINE

● PREDICTED

Laya

Answers typed questions in a single pass: condition, recommended action, urgency, needs human review. Open source (Apache-2.0), runs on the laptop.

🔒 Never generates text. Corroborates the decision, never overrides it.

SYSTEM 2 • EXPLAINER • CLOUD

● AI-EXPLAINED

OpenRouter LLM

Turns the calculated facts into plain language for the operator. May rank only feasible options, and move the spoilage estimate only within a fixed band.

🔒 Never invents a temperature, ETA, cost or quantity.

PLAIN CODE, NOT AI

Spoilage probability

Anomaly detection

Demand forecasting

Optimization, not vibes.

$$\min(C_{\text{transport}} + C_{\text{food-loss}} + C_{\text{delay}})$$

SUBJECT TO

ETA ≤ remaining safe time

Quantity ≤ available capacity

Storage temperature compatible

COLD STORE	ETA	FREE SPACE	FEASIBLE
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WH01 Central Warehouse	18 min	1,200 kg	✓ met
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WH02 North Cold Store	27 min	800 kg	✓ met
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WH03 South Distribution Hub	46 min	600 kg	✓ met
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• OPTIMIZED

DIVERT → WH01

The optimizer picks the store; the model only explains why.

● LIVE DEMO

One truck. One failure. One decision.

T102 is carrying 500 kg of fresh chicken, safe between 0 and 4 °C. Its refrigeration is about to fail.

```
$ make scenario SCENARIO=REFRIGERATION_FAILURE TRUCK=T102
```

Command Center

T102

Model

Optimization

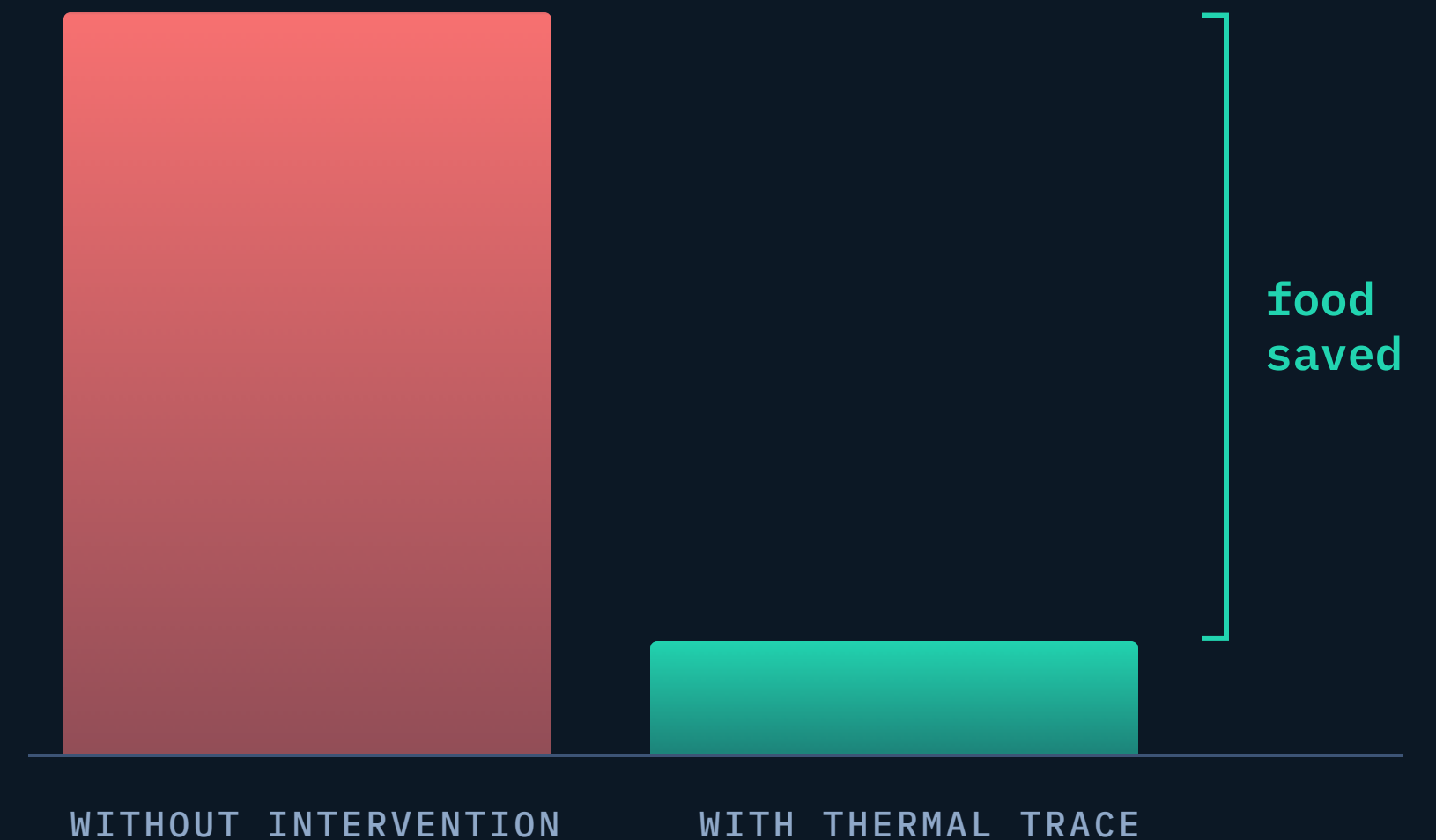
Food loss

Comparison

Food saved. Money not thrown away.

$$\text{food saved} = (\text{loss}_{\text{without}} - \text{loss}_{\text{with}}) \times \text{quantity}$$

In kilograms, and in QAR as financial loss prevented. Computed by the food-loss engine for every intervention. ● CALCULATED



Every value knows **where it came from.**

• MEASURED

Read from a sensor on the truck.

• CALCULATED

Deterministic Python: exposure, deterioration, shelf life, food saved.

• PREDICTED

A model estimate, shown with its probability.

• OPTIMIZED

Chosen by the optimizer under hard constraints.

• AI-EXPLAINED

Plain language about code-computed facts. No new numbers.

○ SYNTHETIC

Simulated for the demo, and labelled as such. Never shown as measured.

Shown on screen, next to the number. Literature values are cited, not certified thresholds.

One operator. One month. Twenty pallets.

On day one we can tell you how much of your loss is **refrigeration** and how much is **scheduling**.

Sensors say what is happening. Mathematics says how much damage has occurred. AI predicts what happens next. Optimization chooses the action. **The food-loss engine proves it saved real food.**

 **Team 2**
REBOOT THE EARTH · CMUQ

01

 Member name

02

 Member name

03

 Member name

04

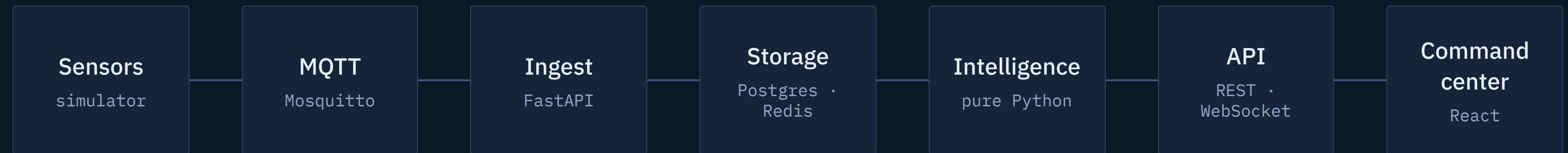
 Member name

05

 Member name

[make demo](#)github.com/Is1-d/Reboot-The-Earth-Cold-Chain-Logistics

Nothing that cannot run on a laptop **in a warehouse office.**



GET `/api/internal/context/{truckId}`

The one contract between the data platform and the intelligence engine, so the two can never disagree about what a truck is experiencing.

0 openly licensed sources, served live.

GEOGRAPHY4 GeoNames places OSM cold-chain sites OSM HGV restrictions Natural Earth	WEATHER3 Open-Meteo forecast Open-Meteo ERA5 NASA POWER	FOOD SCIENCE4 Product storage table USDA Handbook 66 Codex Alimentarius UNECE ATP	EMISSIONS2 UK DESNZ / DEFRA factors EEA / EMEP road transport
ROUTING1 OSRM road routing	IMPACT1 FAO Food Loss & Waste	PHYSICS1 Psychrometrics (Magnus)	OPERATIONAL1 Public holidays

Catalogue at GET /api/opendata. Literature values are cited, not certified thresholds.

Runs anywhere. Proves itself.

make demo

One command

Broker, database, cache, backend, simulator and dashboard. With no API key it still runs end to end; only the prose is plainer.

0

Tests, no network

`make test` runs the pytest suite on SQLite with the LLM mocked; `tests/test_demo.py` replays the whole failure.

Food → Rx

Beyond food, later

Product profiles drive the condition engine, so the same exposure → integrity → risk logic extends to pharmaceuticals and vaccines.